

Project Design Phase-I
Proposed Solution Template

Date	7 March 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Electric vehicles are quickly becoming more common in urban transportation. However, understanding charging behavior, battery performance, and efficiency across different EV models is crucial for energy planning, fleet management, and consumer choices. We need a system that can analyze EV charging data, battery health metrics, and driving range to provide useful insights. A visualization tool can assist city planners, fleet managers, and consumers in making informed decisions.
2.	Idea / Solution description	The proposed solution is a data visualization tool created using Tableau for analytics dashboards and Flask as a web framework to present these dashboards online. The system will examine EV charging patterns, battery performance, and model efficiency. Interactive dashboards will show metrics like charging time, energy consumption, state of charge, predicted vs actual driving range, and efficiency comparisons among EV models. Users can filter data by location, driving style, and time to get tailored insights.
3.	Novelty / Uniqueness	This solution stands out because it combines EV analytics with interactive visualization and a web deployment platform. The system offers dynamic dashboards instead of static reports, allowing users to explore charging trends, range efficiency, and battery performance in real time. By integrating Tableau visualizations with a Flask-based web interface, we ensure easy access and better usability for various stakeholders

4.	Social Impact / Customer Satisfaction	The project promotes sustainable transportation and smart city planning. City planners can better use the electric grid and plan charging infrastructure. Fleet managers can track battery health and lower operational costs by anticipating maintenance needs. Consumers can compare EV models based on efficiency, cost per kilometer, and environmental impact. In summary, the system enhances customer satisfaction and encourages eco-friendly vehicle adoption
5.	Business Model (Revenue Model)	The solution can work as a subscription-based analytics platform for EV fleet operators, energy planners, and charging station providers. Additional revenue can come from premium analytics features, custom dashboards, and consulting services for EV infrastructure planning
6.	Scalability of the Solution	The system is highly scalable as it can connect data from various EV manufacturers, charging stations, and fleet systems. Cloud deployment can enable real-time data processing and visualization for large datasets across several cities or regions. The architecture allows for growth with extra analytics modules and IoT-based vehicle data sources.